Increasing electricity exports through delivery-informed allocation in distribution networks

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Abstract

Delivering surplus electricity from distributed generation depends on how access to a constrained network is shared. We study request-based allocation before output is known, with awards fixed within each interval. A larger request can protect its sender's export while reducing total delivery. An unused award alone does not reveal this loss. We introduce externality-conditioned counterfactual feedback (ECF) to update later request ceilings from completed delivery and post-event estimates of available export. ECF recalculates allocation with the unused-award group's requests reduced to delivered amounts. The allocation rule, loads, available output and network limits remain unchanged. A comparison that increases delivery elsewhere and in total and passes the specified AC checks triggers a smoothed reduction in later ceilings. In the primary controlled comparison across five allocation rules, ECF increases mean delivered power by 24.8 kW relative to proportionally distributing the same total request reduction. An 80-day Australian public-data simulation yields 1.39 MWh (0.59%) more modeled export than no feedback. ECF also delivers 1.34 MWh more than the mean of tested random updates retrospectively calibrated to a similar frequency. These findings support using feasible opportunities to recover delivery, rather than unused permission alone, to guide subsequent allocation.

1 Introduction

Connecting distributed generation is only part of the task of bringing its electricity into use. Rooftop photovoltaics (PV) and larger distributed generators must share a network whose voltage and equipment limits constrain simultaneous exports. Electricity available at one connection can therefore be held back by conditions elsewhere. Network reinforcement and flexible power-flow control can expand the operating range[1]. Within that range, how access is shared also determines which generation can reach the network. China's distributed-PV management framework similarly places connection capacity and operational control alongside project development[2]. The practical task is thus to turn available generation into delivered electricity, both by providing sufficient network capacity and by making effective use of the capacity already available.

Flexible export arrangements provide one way to manage this changing capacity. South Australia's flexible-export programme offers an alternative to restrictive fixed limits[3], and California's limited generation profiles permit export schedules aligned with time-varying hosting capacity[4]. The European Union also provides for flexible connection agreements where network capacity is limited[5]. Although these arrangements use different procedures, each makes permitted export part of ongoing network management. Operating envelopes express this permission as connection-level bounds, allowing owners or aggregators to schedule their resources within network constraints[6–8]. These

bounds let an operator constrain simultaneous power flows without prescribing every device's output. Their energy benefit nevertheless depends on whether the connections receiving permission have electricity available to use it. Alongside calculating safe limits, an operator must therefore consider how subsequent allocation should respond when their use differs from what was anticipated.

A feasible allocation need not be the allocation that delivers the most available electricity. When connections share a binding constraint, increasing one connection's permission can reduce another's. If the first connection cannot use its permission while the second has additional output, some electricity can remain undelivered even though another allocation would respect the same limits. Whether that alternative is feasible depends on the locations and simultaneous injections: unused permission is not a freely transferable block of network capacity. Reducing awards until little permission remains unused is not necessarily an improvement either: a smaller award can prevent useful export when output rises. The coordination problem is to identify where a change in access could deliver more electricity, without treating every unused award as a reason to restrict future output.

Requests provide one way to express intended use before a coordinator allocates feasible access. A related information sequence appears in Project Converge's shaped-operating-envelope design: aggregator bids and customer contribution plans precede the calculation of export bounds[9]. Here we study a request-based arrangement in which an award is an upper bound on export, not an instruction to generate that amount. It remains fixed within the interval, so unused access is not reassigned before the interval ends. Participants request access while their output is still uncertain. A request above the point forecast can preserve permission for high output, whereas requesting only the forecast can forgo that export. Low output can then leave part of the larger award unused. The same request can therefore protect useful generation in one realization and restrict another connection in another. The requester values the export that its extra permission protects, whereas the change in total delivery also depends on who loses access and whether they could have used it. Its consequence cannot be inferred from request size or forecast error alone.

Fig. 1 separates these possibilities. Connections A and B share a binding limit. In (b) and (c), B leaves the same amount of access unused. Reallocating it does nothing in (b) because A has no additional electricity available. In (c), A could use the released access, so total delivery would rise. B's award and metered delivery look alike in the two cases. They reveal unused permission, but not its cost to another exporter. Nor does A's delivery at its limit reveal how much additional output was available behind that limit. Even the awards and deliveries of both connections can therefore be insufficient to identify the lost opportunity.

Existing research provides several ways to improve this allocation. Robust operating envelopes account for uncertain injections when defining network-feasible bounds[10,11], while network-secure bidding combines those bounds with adjustments to uncertain output[12]. Allocation objectives can favour total access, proportional sharing or equality[13–16]. Market mechanisms coordinate bids and prices with network constraints and participant incentives[17–19]. These approaches determine which uses of the network should be admitted using the information available before delivery. Project EDGE's comparison of forecast update frequencies also shows why the timing and accuracy of that information matter for operating envelopes[20]. After delivery, an additional assessment becomes possible: determining whether an alternative feasible allocation would have delivered more of the output that was available.

History-dependent allocation and access exchange are particularly relevant to this question. Fair-over-time coordination spreads curtailment across participants and intervals[21]. Stateful pricing and allocation uses historical delivery relative to available export to adjust subsequent priorities[22]. SecuLEx instead permits network-checked exchange of issued limits according to customer needs[23].

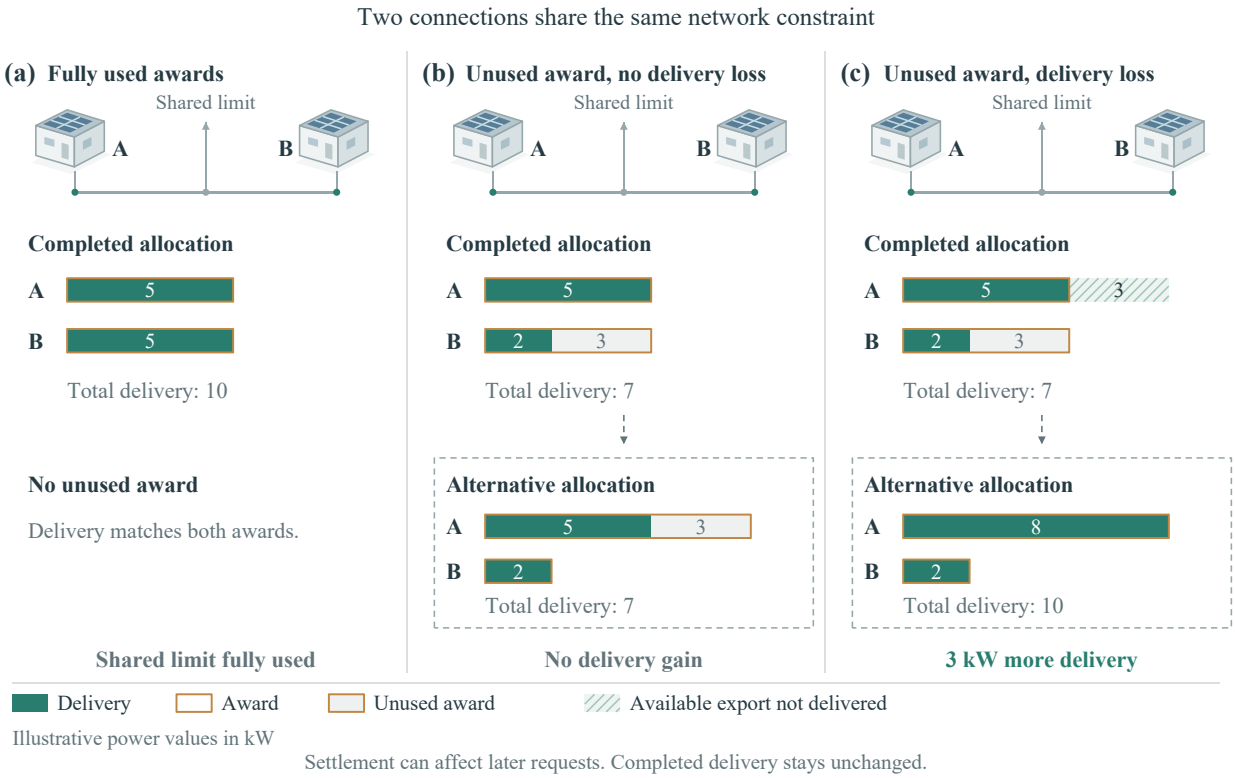


Figure 1: When unused export permission displaces delivery. Two connections share a binding network constraint. (a) Both awards are fully used. (b) B leaves part of its award unused, but A has no additional available export. An alternative allocation increases A's award without increasing delivery. (c) B leaves the same portion unused while A has available export that its award prevents it from delivering. The alternative allocation increases total delivery. Teal fills show delivery, amber outlines show awards, and hatching shows available export not delivered. Numbers within each segment give its power in kW. Dashed frames show calculations for the completed interval. Settlement can affect later requests but cannot restore that interval's lost delivery. The illustrative values use a shared 10 kW authorization bound and initial awards of 5 kW each; they are not case-study results.

These approaches establish ways to redistribute access within or across intervals. The question here is what a completed shortfall should change in a later allocation. An unused award records permission not exercised; unmet available export records electricity not delivered. Neither quantity alone gives the net delivery change that a specified reduction in requests would produce through the existing allocation rule. That change depends on the alternative awards, the output available to use them and the feasibility of the resulting power flows. It must include any delivery lost within the group whose requests are reduced. The information sought is therefore the delivery consequence of a specific reassignment, rather than another measure of an individual shortfall.

This distinction also explains the role of settlement. A charge based only on B's shortfall gives the same signal in Fig. 1(b) and (c), although only one contains recoverable delivery. A charge informed by system-wide delivery effects could use a richer signal; the issue is which information supports it. Faster reassignment can address a mismatch before the interval ends. Where awards remain fixed, completed records instead offer a basis for improving later allocations, consistent with measurement-

based approaches to power-system coordination[24,25]. The resulting task has two parts. One is to identify delivery that a feasible alternative allocation could have recovered in the completed interval. The other is to establish whether using that information improves later delivery. The second does not follow from the first: a restriction based on low output can impede export when output recovers. Useful feedback must therefore distinguish opportunities to recover delivery and retain room for changing generation.

We examine these two questions using one uncertain-output model for both request choice and subsequent delivery. A larger request can increase the sender's expected payoff and unused award while reducing total delivery, which we call a request–access–delivery externality. The same model also produces large requests that protect useful export. The distinction is therefore a consequence of how available power and allocated access coincide across connections, rather than a label attached to every large request.

We then introduce externality-conditioned counterfactual feedback (ECF). Fig. 2 shows its timing. After an interval ends, the allocator recalculates its allocation with the unused-award group's requests reduced to completed delivery. Topology, load, network limits and the allocation rule stay fixed; awards and power flows are recalculated. A post-event estimate of available export tests whether the released access would have increased delivery elsewhere and in total while passing the AC checks. This estimate must describe the export available during the interval without the award constraint and remain valid under the alternative awards; the delivered amount alone is insufficient. A positive test lowers the group's request ceilings for the next interval. Otherwise they recover toward connection bounds. This changes a later input to the existing allocation rule, not a completed award.

The experiments test whether this information makes later allocation more effective, rather than merely describing a past loss. In the primary controlled comparison across five allocation rules, ECF preserves most of the additional delivery obtained by requests above the point forecast. Total export exceeds both no feedback and proportional reductions of the same total request. In the Australian public-data simulation, ECF also exceeds the mean delivery of tested random updates retrospectively calibrated to a similar frequency. Restricting requests more strongly does not continue this improvement: replacing ceilings outright with the latest targets lowers total delivery below no feedback in the controlled zero-cost comparison and the Australian simulation. These findings connect the benefit of selecting restrictions to the need to preserve access for changing output. The contribution is to make completed delivery useful for subsequent sharing of the network, without treating unused permission itself as the loss to eliminate.

2 Results

The practical objective is to deliver more available electricity within the same network limits, rather than to maximize the use of every award. We first test whether a request that helps its sender can reduce total export. If so, a delivery improvement from feedback could still have a simpler explanation: it may merely reduce requests or update ceilings less often. The controls test whether reducing the same total request or updating at a similar frequency reproduces ECF's delivery gain. They examine which connections to restrict and which intervals should trigger an update. We then examine when the ceiling response reverses the delivery gain. Table 2 in Methods summarizes the controls.

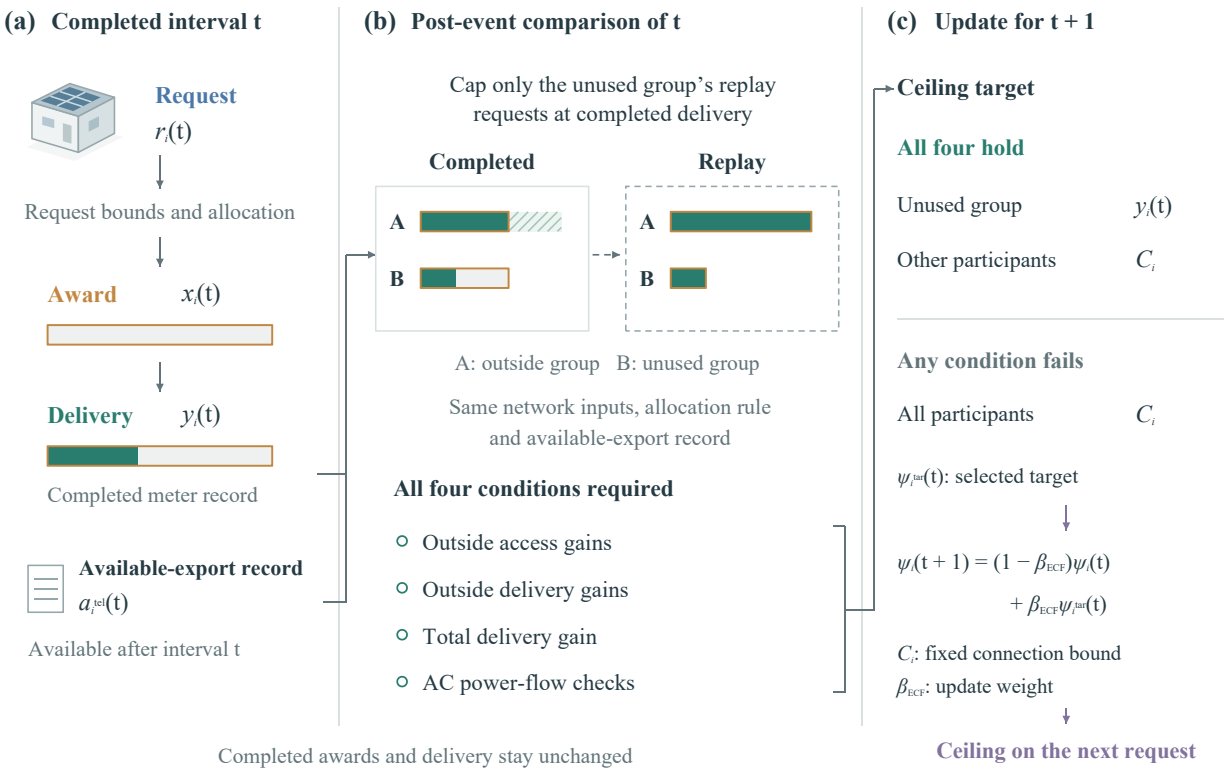


Figure 2: Using completed delivery to update later request ceilings. (a) Submitted requests pass through connection and current feedback bounds before the existing rule allocates access. (b) After delivery, requests from the unused-award group are reduced to completed delivery and allocation is repeated under the same exogenous network conditions. The comparison uses a post-event available-export record and checks whether delivery could rise outside the group and in total without violating the specified AC limits. (c) A qualifying comparison sets a lower target for the group's next request ceiling; otherwise, ceilings recover toward connection bounds. The completed award and delivery do not change.

2.1 Private delivery gains can accompany a larger loss elsewhere

In 1,500 controlled parameter–location cases without a shortfall cost, allowing a participant to choose a larger request raises its expected delivery by 5.2 kW on average relative to a point-forecast request. Other participants lose 18.8 kW, so total delivery falls by 13.5 kW. The requester gains useful export, but the access it receives displaces more export elsewhere. Both outcomes follow from the same model of uncertain available power: there is no need to assume that a large request is unproductive to obtain a loss in total delivery.

Fig. 3 traces the request choice as forecast uncertainty σ and normalized shortfall cost ϑ change. The cost parameter is the modeled charge for one unit of unused award divided by the value of one unit delivered. We test five allocation rules—weighted proportional (WP), equal reduction (ER), equal access (EA), maximum export (ME), and fairness-aware allocation (FA)—under two reactive-power settings. Each larger request is compared with the same participant's forecast-based request; Methods specifies the output model.

The mean loss does not describe every larger request. We classify a request–access–delivery ex-

ternality when the requester's expected payoff and unused award rise while total delivery falls. A productive large request raises payoff without a total-delivery loss beyond the numerical tolerance. This classification concerns the effect of choosing a larger request; ECF instead tests whether a completed unused award could have supported delivery elsewhere. Methods and Supplementary Section S2 define the two comparisons.

At $\vartheta = 0$, 71.5% of the 1,500 cases meet the externality definition, while 11.9% are productive large requests. The other 16.6% either retain the forecast-based request or meet neither classification. Thus request size and unused access alone cannot identify the delivery outcome. These shares pool all five uncertainty levels; Fig. 3(c) fixes $\sigma = 0.20$ to show how the categories vary with shortfall cost.

Uncertainty makes the higher-output outcome worth protecting; a higher shortfall cost instead discourages unused awards. At $\vartheta = 10$, 10.7% of cases still meet the externality definition, and the mean total-delivery loss falls to 0.6 kW across all cases. Productive large requests still account for 11.5%. Thus a charge can reduce delivery displacement, but the relation between request size and productive export still depends on the available output.

Panel (d) shows why a single shortfall cost cannot represent every loss to others. For each 0.1 increase in the request multiplier, it divides the additional outside-delivery loss by the additional unused award. Across 4,608 steps that increase own delivery and unused access while reducing total delivery, the median ratio $\vartheta^{\mathrm{ext}}_{i,\Delta}$ is 2.29 (interquartile range 1.10–5.72). Rule-specific medians range from 0.92 for ME to 5.37 for EA. This variation explains why the amount left unused is not a uniform measure of its cost to other exporters. It motivates checking the delivery opportunity after an interval, alongside any advance incentive. The next comparison asks whether later request restrictions can preserve the private delivery protected by larger requests while reducing the greater loss elsewhere. Supplementary Section S2 derives the ratio and its domain.

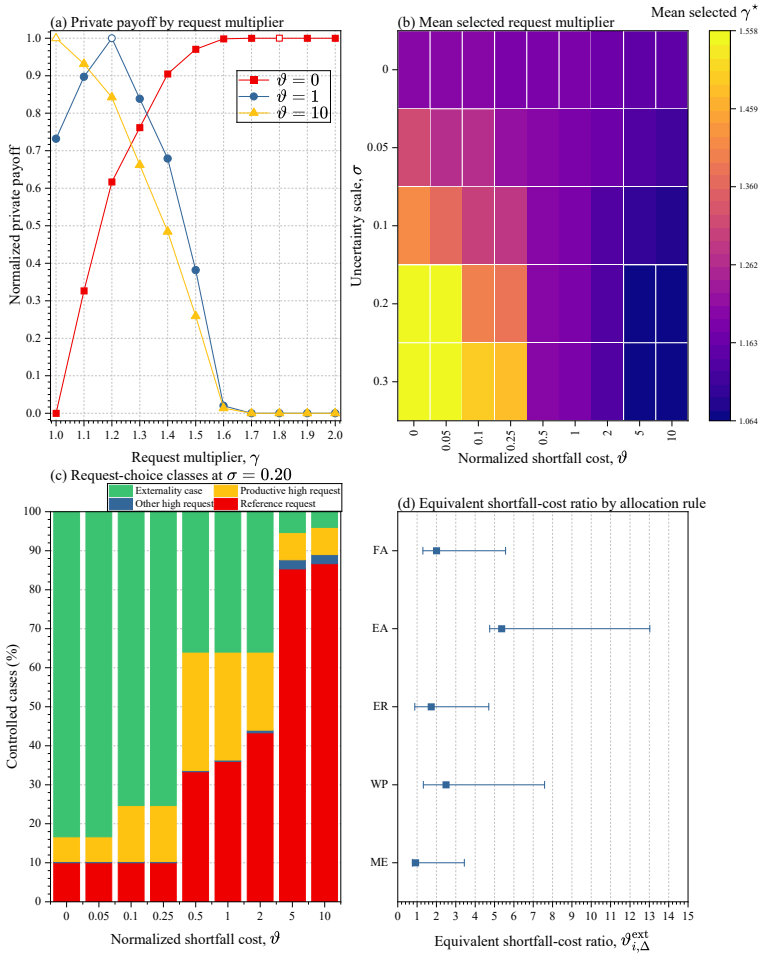


Figure 3: A larger request can protect own delivery or reduce total delivery. (a) Private-payoff curves for participant 21 under WP, Q0 and $\sigma = 0.20$, at three normalized shortfall-cost levels. Within each curve, payoff is rescaled as $(U - U_{\min})/(U_{\max} - U_{\min})$; a zero span is plotted as zero. (b) Mean selected request multiplier for each uncertainty and shortfall-cost setting. Means cover participants, rules and reactive-power settings. (c) Shares at $\sigma = 0.20$ ($n = 300$): reference requests; productive high requests, which raise payoff without a total-delivery loss beyond the classification tolerance; request–access–delivery externality cases, which raise payoff and unused awards while lowering total delivery; and other high requests, which meet neither high-request definition. (d) Finite-difference cost ratio $\vartheta^{\mathrm{ext}}_{i,\Delta}$: expected delivery loss elsewhere per unit increase in expected unused award for a 0.1 request-multiplier step. Squares show rule-specific medians, and bars show interquartile ranges. WP, ER, EA, ME, and FA denote weighted proportional, equal reduction, equal access, maximum export, and fairness-aware allocation, respectively.

2.2 Targeted request reductions preserve useful export access

Across 16-interval output trajectories, ECF delivers 24.8 kW more on average than a control that removes the same total request across all participants. This comparison uses zero shortfall cost and an intermediate feedback gain, $\beta_{\mathrm{ECF}} = 0.5$; only the placement of the reduction differs. We also compare a charge before allocation with feedback based solely on each participant's own delivery. Feedback comparisons share raw requests and network conditions; the shortfall-cost comparison

changes the initial request choice. Supplementary Fig. 1 follows one trajectory as output falls and recovers.

2.2.1 A shortfall cost reduces displacement but also useful delivery

A shortfall cost changes requests before anyone receives an award. Raising ϑ from 0 to 1 reduces the mean multiplier from 1.51 to 1.14 and the share of intervals meeting the economic externality definition from 50.1% to 15.6%. But it also removes 45.6% of the larger-request group's delivery gain (Supplementary Fig. 2). At $\vartheta = 10$, none of these intervals meets the externality definition, while the group's gain is almost zero. The charge reduces the modeled conflict partly by removing useful high-output access as well. This economic classification is not the replay test that activates ECF.

At $\vartheta = 0$, ECF improves total delivery by 24.9 kW over no feedback while retaining 79.4% of the larger-request group's original delivery gain. This recovery changes the comparison with forecast-based requests from a 21.7-kW loss without feedback to a 3.2-kW gain with ECF. Mismatch-only feedback, which updates each ceiling from that participant's delivery-to-award ratio, recovers 15.2 kW relative to no feedback. ECF therefore delivers 9.7 kW more than this shortfall-based update. Supplementary Section S3 reports the group decomposition and other cost settings. At $\vartheta = 10$, ECF raises mean delivery but increases the economic externality share from zero to 1.3%. Its completed-interval test therefore does not minimize that separate request-effect classification.

2.2.2 Where requests are reduced determines recovered delivery

A gain over no feedback does not by itself show that the distribution of request reductions matters. The equal-total control removes the same total request as ECF in each interval but distributes the reduction proportionally across connections. ECF's 24.8-kW mean advantage at $\vartheta = 0$ therefore isolates the value of placing the reduction. ECF delivers more than this control in 95.6% of 90 matched trajectories and more than mismatch-only feedback in 73.3%. Its mean advantage over both remains positive at $\vartheta = 1$ and 10.

Selecting a useful interval is insufficient if the next ceiling is too tight. At $\vartheta = 0$, total delivery lies 2.8 kW below the forecast-based reference at $\beta_{\mathrm{ECF}} = 0.25$, rises 3.2 kW above it at 0.5, and falls 23.4 kW below it at 1 (Fig. 4). The reversal also occurs at the other shortfall-cost settings. A unit-gain update replaces the previous ceiling with one interval's target, which can restrict export when output subsequently recovers. The intermediate gain is the primary comparison setting and gives the largest mean delivery among these three settings.

The controlled trajectories create contested access by design. The replay therefore qualifies in 12,910 of 12,960 attempts, with a median 37.8-kW potential recovery in the completed interval. These trajectories chiefly test targeting and update gain. The public-data simulation later tests interval selection under more variable conditions: only 12.6% of its half-hours qualify. This separates two requirements for useful feedback: placing a reduction where it can release usable access, and deciding when a completed shortfall warrants that reduction.

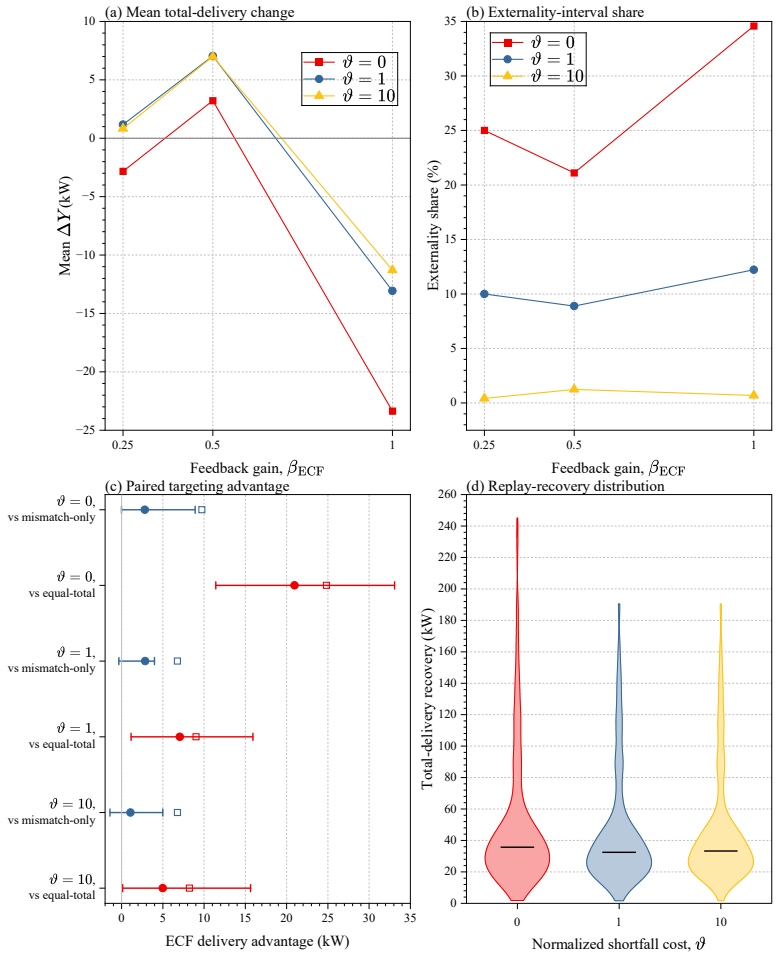


Figure 4: Targeted feedback preserves delivery, but a unit-gain update reverses the improvement in the controlled cases. (a) Mean signed change in total delivery relative to forecast-based reference requests and (b) externality-interval share for the three tested feedback gains. The series denote $\vartheta = 0, 1, 10$. (c) ECF delivery advantage in matched trajectories. Rows labelled "vs mismatch-only" compare ECF with mismatch-only feedback. Rows labelled "vs equal-total" compare ECF with the equal-total control. Panel (c) uses the primary gain, $\beta_{\mathrm{ECF}} = 0.5$. Filled circles show medians, open squares show means, and whiskers show interquartile ranges. The gray line marks zero. (d) Distribution of total delivery recovered when the replay meets all four conditions. Violin width shows empirical density, and the black horizontal segment shows the median.

2.3 Output timing and allocation rules alter the delivery gain

The equal-total comparison identifies a benefit from directing request reductions, but the allocation rule and subsequent output determine how much released access can be used. We therefore compare rules and output patterns, together with the number and placement of changed requests (Fig. 5). At $\vartheta = 0$ and $\beta_{\mathrm{ECF}} = 0.5$, mean ECF delivery exceeds both no feedback and equal-total reduction in every factor-level average shown. The advantage over no feedback ranges from 8.9 kW under FA to 67.5 kW under EA; against equal-total reduction it ranges from 9.7 to 50.8 kW. The effect is not specific to one of the five rules, although its size depends on the rule.

The three temporal profiles preserve each participant's available-export values and frequencies. They

change their order and, because participants have different phase offsets, which output states occur together across connections. ECF gains 35.3 kW over no feedback when low-output intervals are clustered, 12.2 kW when they are intermittent, and 27.3 kW in a shock–recovery sequence. The respective gains over equal-total reduction are 34.4, 12.6 and 27.4 kW. The response to past delivery therefore depends on the temporal scenario, including both persistence and coincidence of output across connections.

When one participant changes its request, the mean gain over no feedback is 11.2 kW. It is 35.0 kW for five nearby participants and 28.6 kW for five dispersed participants. Each chooses its own request multiplier. The latter two cases change participant identity as well as electrical placement, so their difference cannot be attributed to distance alone. Zero participant reactive power (Q0) and 0.95-power-factor absorption (Q95) both give a 24.9-kW mean gain over no feedback at the reported precision, and 24.8 kW over equal-total reduction.

2.3.1 AC feasibility and allocation equality

The controlled award and delivery states satisfy nonlinear AC convergence, power-balance, voltage and two-ended apparent-power criteria within the declared synthetic operating envelope. Maximum branch utilization is 90.91% of the scenario rating. Supplementary Fig. 3 and Section S3.5 report the voltage margins and full coverage of these checks.

More delivery need not mean more equal access. The capacity-normalized Jain index in Supplementary Fig. 3 measures equality of awards, not social fairness. At $\vartheta = 0$, ECF improves its mean change relative to no feedback under all five rules, but four remain below the forecast-based reference. EA moves from -2.4 percentage points without feedback to 0.1 with ECF; WP, ER, ME and FA remain between approximately -0.8 and -0.5 points. ECF also exceeds equal-total reduction on this mean index for every rule. The additional authorization and unused-award results appear in the Supplementary Information.

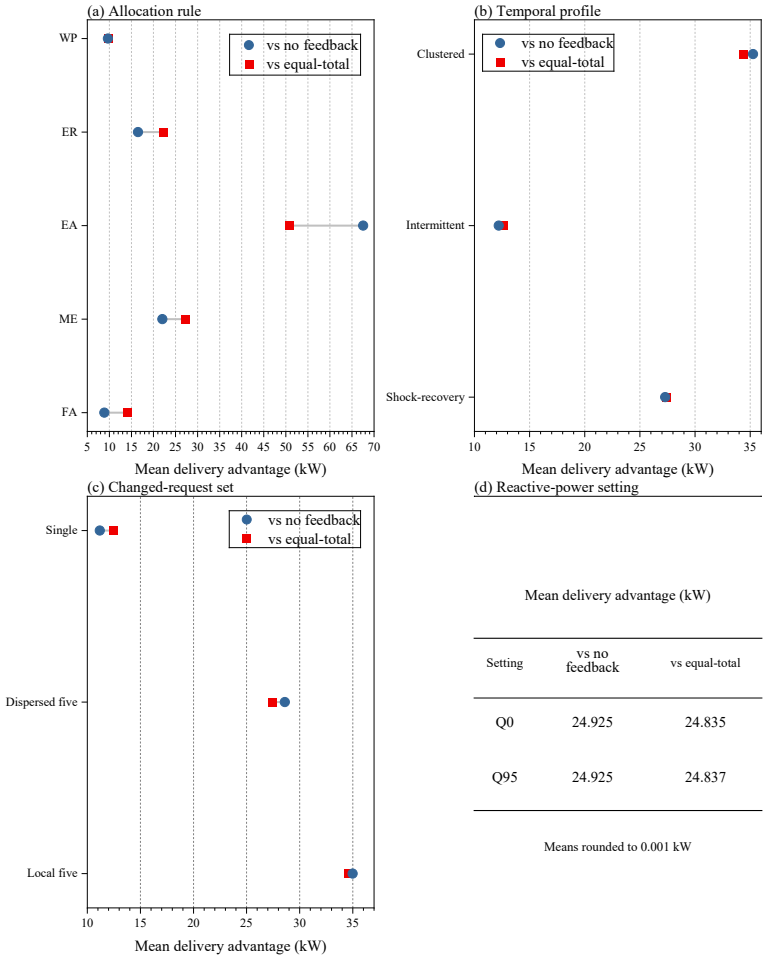


Figure 5: The controlled delivery advantage persists across allocation rules and output timing at $\vartheta = 0$ and $\beta_{\mathrm{ECF}} = 0.5$. Markers are factor-level means, grouped by (a) allocation rule, (b) temporal profile and (c) changed-request set. Panel (d) tabulates means for the two reactive-power settings to show their close agreement. Blue circles compare ECF with no feedback. Red squares compare ECF with the equal-total control. WP, ER, EA, ME, and FA denote weighted proportional, equal reduction, equal access, maximum export, and fairness-aware allocation, respectively. Q0 sets participant reactive power to zero, whereas Q95 uses 0.95-power-factor absorption. The temporal profiles preserve individual output frequencies but change ordering and cross-connection coincidence.

2.4 Public-data simulations show when targeting raises export

The controlled trajectories show a delivery gain across rules, but create repeated opportunities to reassign access. We next use household generation and demand histories to test whether feedback remains useful when these opportunities occur irregularly. An unused award need not coincide with export held back elsewhere. The comparisons therefore examine both the energy gained over time and the value of choosing when to update.

2.4.1 Better use of access increases delivery, not just utilization

Across 80 Australian network-days, ECF delivers 239.36 MWh of modeled export: 1.39 MWh (0.59%) more than either no feedback or equal-total request reduction (Fig. 6). The simulation combines a published medium- and low-voltage network with public household PV and load histories, from which requests and available export are constructed. The primary comparison uses proportional allocation and initializes ceilings each day; Methods describes the profile mapping and day selection.

The gain combines more authorization with less unused permission. Relative to no feedback, ECF authorizes an additional 1,206 kWh and leaves 186 kWh less unused. Together these changes account for the additional 1,392 kWh delivered, according to the identity in (8). Equal-total reduction has almost no delivery effect because proportional scaling preserves request shares: a binding network limit can be met by changing the common allocation factor instead. Targeted feedback changes those shares and therefore the allocation of a binding network limit. Supplementary Sections S4 and S11.1 give the derivation and full energy balance.

The gain is not uniform across days. ECF delivers more on 44, less on 32 and the same amount on four, relative to either control. Its mean gain is 17.4 kWh per assessed day; the dependence-adjusted 95% intervals are 5.1–29.7 kWh against no feedback and 4.9–29.9 kWh against equal-total reduction. A continuous 35-day block contributes 1.06 MWh of the 1.39-MWh gain. The other 45 days contribute 0.33 MWh, with an interval that includes zero when considered alone (Supplementary Section S10.3). ECF selects a delivery-target reduction in 483 of 3,840 half-hours (12.6%). Previously tightened ceilings can remain below capacity between these selections.

The additional connection-point delivery reduces net upstream import by 1.38 MWh relative to either control, about 98.9% of the connection-point gain. The feeder remains a net importer; Supplementary Section S11.1 gives the circuit loss and passive-element terms. By contrast, previous-day persistence uses 89.0% of its authorization but delivers 50.33 MWh less than ECF. It omits the forecast increment and therefore requests too little on many high-output intervals. Utilization alone rewards a small, fully used award even when more electricity could have been delivered.

The public-study trajectories satisfy their specified allocation and delivery criteria; Supplementary Section S14.3 collects the physical-check coverage for the primary comparison and subsequent tests. Network limits reduce the proportional allocation factor in 44.4% of ECF half-hours; the common network constraints remain active even as the distribution of requests changes.

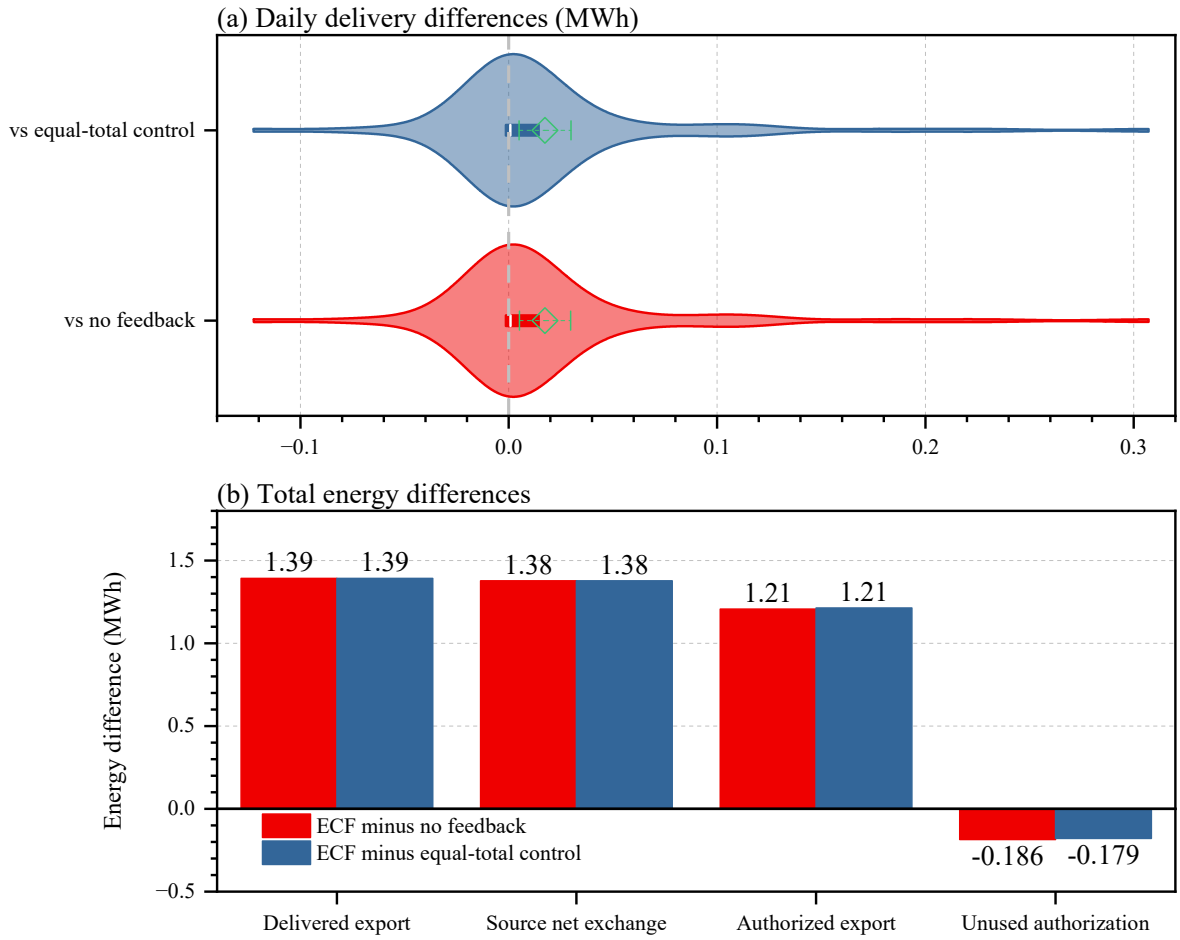


Figure 6: Modeled delivery gains and their energy balance in the Australian public-data simulation. Available export is constructed from PV and load profiles. (a) Paired daily delivery differences for ECF against no feedback and the equal-total control. Violin width shows empirical density. Dark boxes span the 25th–75th percentiles, and white ticks show medians. Green diamonds show means; capped green lines show 95% intervals adjusted for short-range dependence between neighboring dates. (b) Aggregate 80-day differences in delivered export, source-terminal net exchange, authorization, and unused authorization. Each difference is ECF minus the indicated control. Delivery equals authorization minus its unused portion. Here the delivery gain accompanies increased authorization and reduced unused authorization. Positive source-terminal differences indicate reduced net upstream import over the assessed period.

2.4.2 Replay qualification preserves delivery relative to unqualified updates

Only 12.6% of intervals qualify, so the next question is whether the replay adds information or merely reduces the number of updates. We first remove the replay test but keep the same ceiling formula and gain. Updating after every unused award lowers modeled delivery from 239.36 to 232.93 MWh. ECF delivers 6.42 MWh (2.76%) more, with a positive difference on 75 of 80 days. The unqualified rule leaves 18.11 MWh less authorization unused, yet grants 24.54 MWh less authorization. It removes useful export along with idle awards.

That comparison also changes update frequency: 3,086 half-hours rather than ECF's 483. A second

control keeps the target and gain but selects unused-award intervals without replay. Its selection probability is calibrated retrospectively to ECF's qualifying fraction among unused-award intervals. The 12 fixed schedules select delivery targets in 463–519 half-hours. Every schedule delivers less than ECF, by 0.92–1.60 MWh. On average they deliver only 0.05 MWh more than no feedback, versus ECF's 1.39 MWh gain. They also leave almost the same authorization unused as ECF (90.345 versus 90.341 MWh). After averaging schedules within each date, ECF gains 1.34 MWh, or 16.8 kWh per assessed day. The difference is positive on 66 of 80 days, with a dependence-adjusted 95% interval of 7.8–25.8 kWh per day. The tested random schedules yield less delivery at a similar selection frequency, supporting replay-based interval selection within the feedback loop. Supplementary Sections S11.2–S11.3 give the full trajectories and design.

Table 1 collects the corresponding 80-day energy totals for each update rule.

Table 1: Modeled export over the same 80 Australian network days. The unqualified update uses ECF's ceiling formula and gain after every unused award. The frequency-calibrated random row averages 12 replay-free schedules with the same formula and gain. The equal-total control matches ECF's request reduction in each half-hour. All energy totals are in MWh and are rounded independently.

Control	Authorized	Unused	Delivered
No feedback	328.490	90.526	237.963
Equal-total reduction	328.483	90.519	237.964
Update without replay test	305.159	72.226	232.933
Frequency-calibrated random update	328.357	90.345	238.012
ECF	329.696	90.341	239.356

Fig. 7 shows how the advantage over similarly frequent replay-free updates is distributed across days and control trajectories.

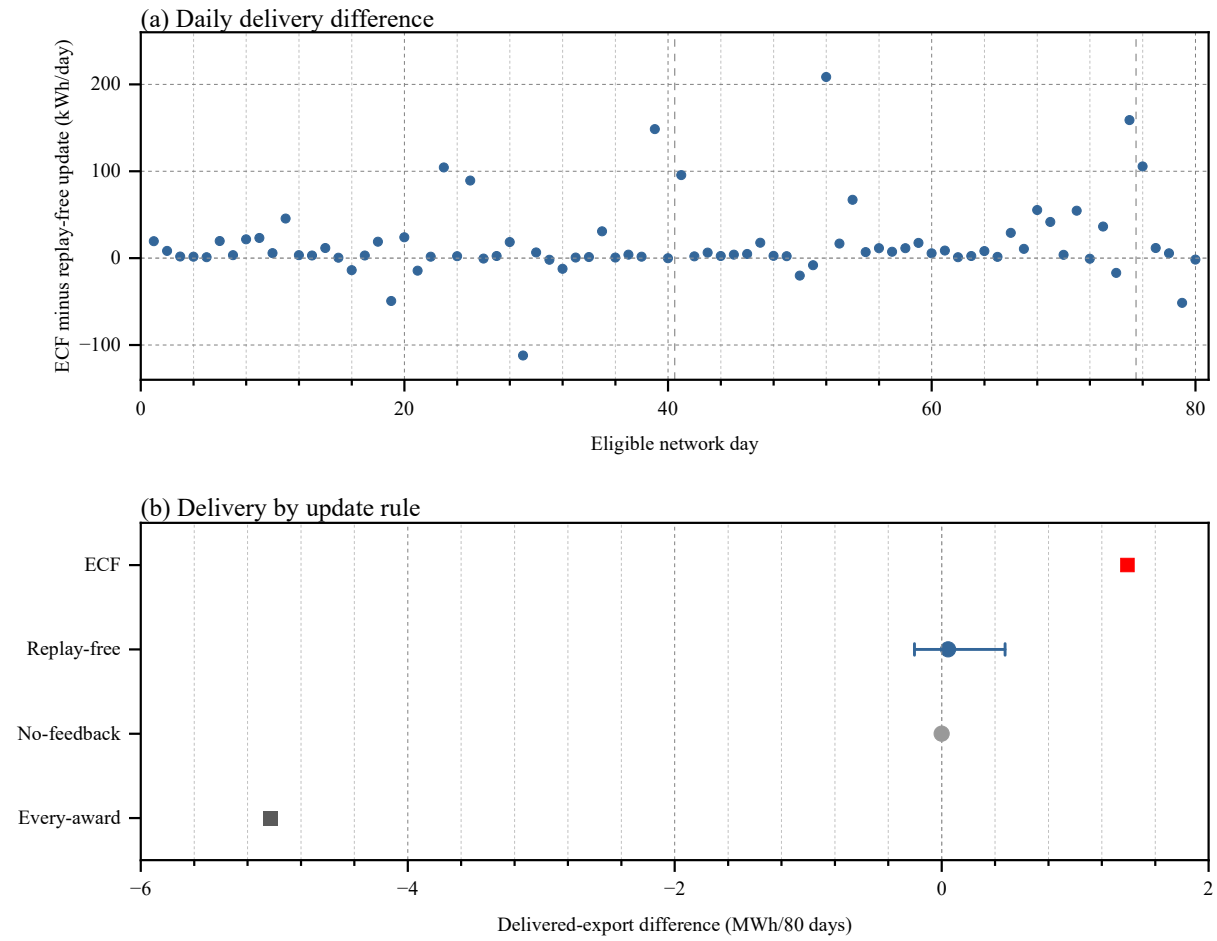


Figure 7: Replay-selected updates recover more modeled delivery than comparably frequent replay-free updates in the Australian simulation. (a) Paired daily modeled-delivery difference between ECF and the mean of 12 frequency-calibrated random updates. The vertical dashed lines bound the longest continuous 35-day block. (b) Eighty-day delivery relative to no feedback. The blue marker is the mean across 12 fixed replay-free schedules; its horizontal range spans their individual totals. The red square marks ECF; gray markers denote no feedback and updating after every unused award. The number of half-hours selected for a delivery-target update is 483 for ECF, 463–519 for the random schedules, 0 for no feedback and 3,086 for updating after every unused award. Ceiling recovery still operates in other intervals. The schedules are retrospectively calibrated to approximately ECF's update frequency, without using replay to choose their intervals. Their independent ceiling histories need not give identical update magnitudes or durations.

2.4.3 Update strength and record error can limit the benefit

Selecting an interval is only part of feedback: the replay concerns completed output, whereas its ceiling applies to later output. At $\beta_{\mathrm{ECF}} = 0.25$, ECF gains 1.34 MWh over each of no feedback and equal-total reduction. At unit gain, it loses 3.04 MWh relative to no feedback: replacing the ceiling with one interval's target can deny export after output recovers. This adverse result remains when ceilings carry across midnight. In ten consecutive-date blocks covering the 80 assessed days, total delivery at each tested gain differs from daily initialization by less than 0.1 kWh (Supplementary Section S12.2).

Imperfect available-export records do not reverse the mean delivery gain in the tested perturbations, but they can change individual replay decisions. We scale the record by ±10%, delay it by one half-hour, or substitute modeled delivery for 10% of records. ECF retains a positive mean advantage over both controls (Fig. 8): 16.3–18.0 kWh per assessed day, with dependence-adjusted 95% intervals above zero.

A separate test adds or subtracts a prescribed error radius of up to 10% of connection capacity. With positive bias, subtracting that radius recovers the unperturbed availability by construction and avoids unsupported qualifications. With negative bias, the lower bound becomes more restrictive: at the largest radius it misses 258 of 496 feasible recoveries along its own trajectory and delivers 269 kWh less than the point-record controller. Both still improve on no feedback (Supplementary Section S12.3). A valid error bound can protect the recovery comparison, but conservative records can also discard useful updates.

Fig. 8 shows these gain and record responses alongside a broader customer-and-calendar comparison, discussed below.

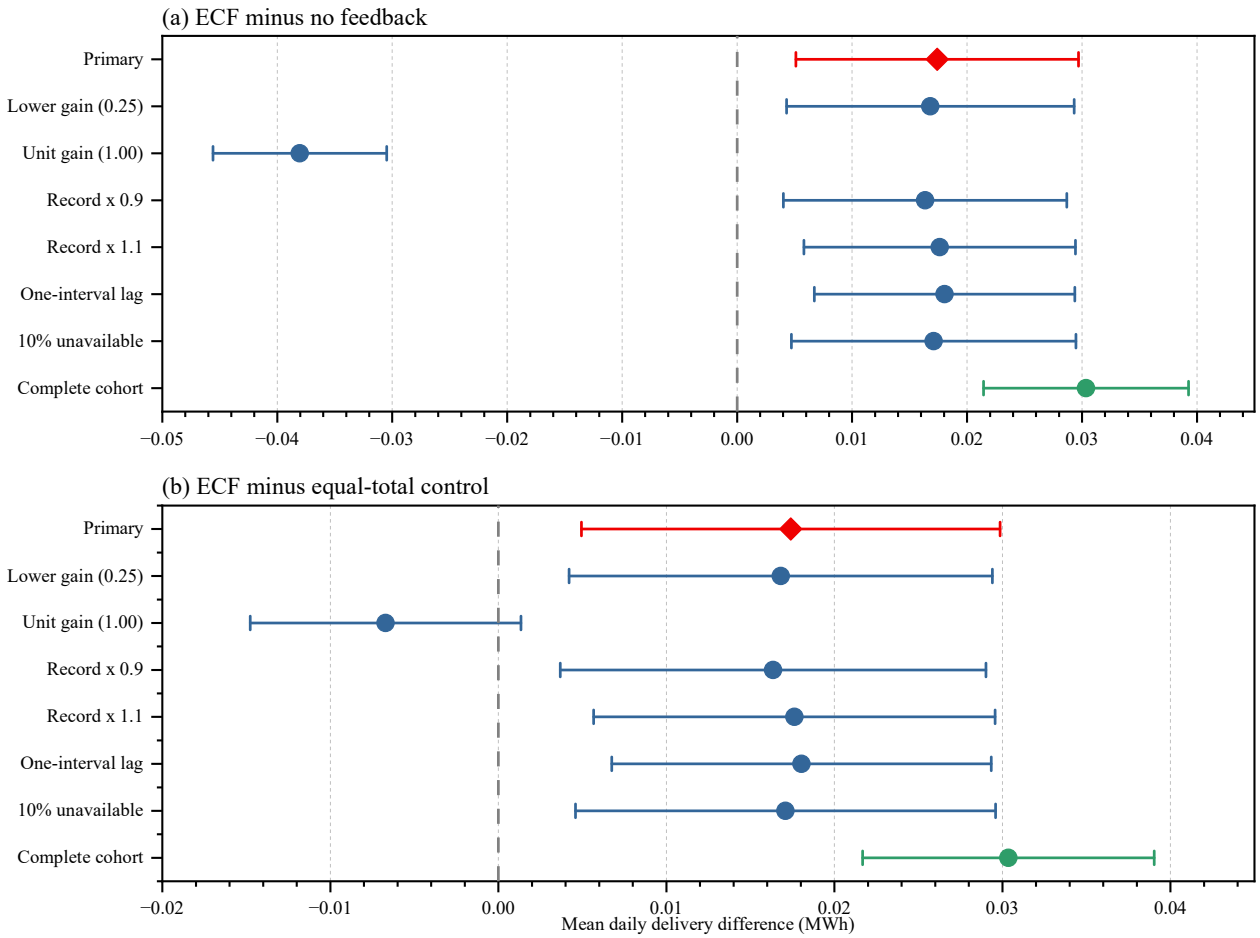


Figure 8: Gain, record quality and customer coverage change the Australian delivery result. Markers show paired mean delivery differences per network day. Bars show 95% intervals adjusted for short-range dependence between neighboring dates. (a) ECF minus no feedback. (b) ECF minus the equal-total control. The red diamond denotes the primary unperturbed-proxy result, and blue circles denote gain or record changes. Green markers denote the complete-customer cohort, which uses the same load multiplier but changes customer composition and calendar coverage (361 days rather than 80). Every transformed available-export record is floored at modeled delivery. The zero line separates positive from negative mean delivery differences.

2.4.4 Spatial placement and available generation shape the benefit

Even with accurate records and a moderate update, recovery requires available electricity at connections that can use additional access. We therefore vary the profile placement, available generation and network, and examine a broader customer cohort.

Changing the placement of household profiles preserves the positive delivery contrast but changes its size. Across five alternative placements, ECF gains about 0.9–5.1 MWh over either control, with positive pointwise mean-difference intervals in all five. These comparisons retain the same 80 days and profiles. They address the constructed spatial mapping, which is necessary because the household and network sources have no common customer identifier. Five capacity scenarios scale both PV generation and connection bounds from half to one-and-a-half times the primary level. The gain is 1.39–1.88 MWh. Placement and available generation change the size of the effect, but neither reverses its sign in these tests. The gain does not scale monotonically with installed PV (Supplementary Sections S13.1–S13.2).

Changing the network itself reveals both gains and null outcomes. In 13 LV sections of the parent feeder, ECF delivers more than equal-total reduction in every section and more than no feedback in 12; LV26 loses 3.7 kWh against no feedback. Among nine representative published LV models, six have qualifying intervals and positive aggregate delivery differences, including a near-zero gain of 0.1 kWh in model M, while three never qualify and remain unchanged (Supplementary Sections S13.3–S13.4 and Supplementary Fig. 5). The variation links the delivery opportunity to the network and the locations of available output.

The complete-customer cohort broadens the calendar to 361 days at the same load multiplier of 0.30. ECF gains 10.96 MWh over each of no feedback and equal-total reduction, or 30.4 kWh per network day; both mean-difference intervals remain above zero. This comparison jointly changes customer composition and calendar coverage (Supplementary Sections S10.3 and S12.1).

The aggregate gain is not shared equally. Among 418 participants in the primary feeder, 330 deliver more than without feedback over the 80 days and 88 deliver less; the largest individual loss is 7.3 kWh (Supplementary Section S14.1). Recovering more electricity in total therefore still leaves a distributional choice about who receives access.

The preceding public-data comparisons construct requests using a fixed forecasting rule. We next allow participants to adjust request-choice probabilities from their own delivery histories. ECF delivers 1.44 MWh more than no feedback and 4.53 MWh more than the same ceiling update without qualification. Each control develops its own request history through experience-based request adjustment, specified in Supplementary Section S14.2.

Across these comparisons, an opportunity to recover export depends on where electricity is available and whether later output can still be delivered under a tighter ceiling. More installed PV alone does not determine the gain. The individual outcomes also show why total delivery and the distribution of access must be assessed separately when applying feedback.

3 Discussion

Completed delivery reveals how much of an award was used, but not how much electricity a different allocation could have delivered. Combining available-export information with network-constrained reallocation makes the latter comparison explicit. The request model explains why this distinction

matters: permission that protects one generator's uncertain output can displace more delivery elsewhere. The feedback experiments show that using information about this opportunity can increase subsequent export while retaining the allocation rule and network limits. The contribution is thus a way to use completed outcomes in later allocation, not merely to describe unused permission after delivery.

ECF uses a feasible alternative allocation to distinguish an unused award from a recoverable loss of delivery. Its value as feedback is established by subsequent export, not by the recovery calculated for the past interval. The equal-total comparison holds the removed request amount fixed and shows that its distribution across connections affects delivery. The retrospectively frequency-calibrated random comparison shows that updating at approximately the same frequency does not reproduce ECF's gain. Together, these controls support the use of available export elsewhere to select restrictions, beyond simply making requests smaller or updating them less often. They also separate this contribution from the choice of the underlying allocation rule, which remains unchanged within each comparison.

The comparisons also change how unused authorization should be interpreted. Updating after every unused award leaves less authorization unused but delivers less electricity than ECF. The mean across the tested random schedules shows almost the same unused amount as ECF, yet less delivery. Low unused authorization can therefore accompany lost export, while similar unused totals can conceal different delivery outcomes. An assessment based only on award utilization would miss these differences. In the primary Australian simulation, the delivery gain is 0.59% over no feedback, and almost all of the added connection-point export appears as reduced upstream import. This establishes an energy consequence of the allocation change at the feeder boundary. Its practical value depends on the opportunities present and the cost of obtaining and processing the required information.

For a coordinator authorized to revise request ceilings, an unused award is a reason to examine the allocation rather than a sufficient reason to reduce later access. The examination must establish whether other connections had additional electricity available, could receive additional awards, and could deliver that electricity within the network limits. The net comparison also accounts for any delivery lost in the updated group. Subsequent performance should be assessed through delivered electricity and its distribution among participants, alongside issued limits and their utilization. These quantities distinguish more effective sharing from restrictions that merely leave less permission to use. Such an assessment can inform request management alongside forecasting, settlement and within-interval reassignment, where the latter is available.

A completed-interval comparison and the response to it solve different problems. At unit gain, a qualifying update replaces the ceiling outright with completed delivery instead of averaging the old ceiling and that target. This produces less total delivery than no feedback in the controlled zero-cost comparison and the Australian simulation. An accurate diagnosis of a past opportunity is therefore insufficient to justify an equally strong restriction on future output. Record quality affects the diagnosis itself: a conservative estimate can avoid unsupported recovery claims while missing opportunities. Individual consequences require separate attention because the comparison tests the joint change in a group's requests. Some participants lose delivery even when the total rises; the group result neither assigns individual responsibility nor guarantees a benefit at every connection. The experience-based request test preserves the mean advantage, but does not model strategic anticipation of future ceilings or manipulation of availability records. The tested gains establish a response trade-off, not an optimal gain for other operating conditions.

The physical and information requirements differ between the two studies. The controlled 141-bus case verifies nonlinear AC voltage, balance and two-ended apparent power under synthetic branch ratings. The Australian models provide measured household histories and published network structures; requests and available export are constructed, and the physical screen covers customer voltage and native transformer ratings rather than line ampacities. An operational study would link requests, awards, metered delivery and verified available export for the same connections and intervals. Availability changes caused by storage, inverter control or flexible demand would also need to enter the alternative-allocation calculation. Applying the update would also require authority to change later request ceilings within the relevant connection arrangement. A study using linked records could assess it alongside that service's forecasting, pricing and within-interval control, and establish whether recovered export justifies the measurement and computation costs.

Completed-interval information can improve the electricity delivered under an existing export-access arrangement. The useful information is not simply that an award went unused, but that a specified, network-feasible change could have enabled more delivery. The simulations show that using this distinction to select later request restrictions can increase export. They also show that an overly strong update can reverse the gain. Managing access therefore involves both identifying a recoverable delivery opportunity and preserving room for later output. ECF connects these tasks without replacing the allocation rule. This provides a way to deliver more available generation within existing network limits, with success judged by electricity delivered rather than permission left unused.

4 Methods

4.1 Requests, awards and available export

The definitions below distinguish the request submitted before allocation, the award that bounds delivery, and the available export used to evaluate an alternative award. The ceiling state links a completed interval to the next admitted request.

Let $\mathcal{N}$ be the participating export connections and $\mathcal{G} = (\mathcal{B}, \mathcal{L})$ the distribution network, with buses $\mathcal{B}$ and branches $\mathcal{L}$. In interval t, participant i requests $r_i(t)$ before its available export $a_i(t)$ is known. Its fixed connection bound is C_i. Feedback changes a request ceiling $\psi_i(t) \in [0, C_i]$, initially C_i, so the admitted request is

$$r_i^{\mathrm{adm}}(t) = \min\{r_i(t), C_i, \psi_i(t)\}. \tag{1}$$

Allocation rule $\mathcal{R}$ then grants access $\mathbf{x}(t) = \Pi_{\mathcal{R}}(\mathbf{r}^{\mathrm{adm}}(t); \mathcal{F}_{\mathcal{M}}(\mathcal{G}, \mathbf{z}(t)), \boldsymbol{\eta}_{\mathcal{R}})$. Here $\mathcal{M}$ denotes the network model, and $\mathcal{F}_{\mathcal{M}}$ its equations and limits under exogenous data $\mathbf{z}(t)$; $\boldsymbol{\eta}_{\mathcal{R}}$ contains rule parameters, held fixed in the comparisons. Completed delivery is $y_i(t) = \min\{x_i(t), a_i(t)\}$, with $0 \le y_i(t) \le x_i(t) \le r_i^{\mathrm{adm}}(t) \le \psi_i(t) \le C_i$. Thus $x_i - y_i$ is the unused award. Awards remain fixed within an interval; access is not redistributed during that interval. Feedback modifies only the admitted-request input in a later interval, not the rule or network limits. These variables are interval-average powers. Multiplication by interval duration and summation give the reported energy totals. Supplementary Table 1 collects the quantities.

The coordinator does not observe participants' private output signals or conditional distributions before allocation. It observes completed delivery and uses a post-event available-export record $a_i^{\mathrm{tel}}(t)$ for the replay. The controlled experiment supplies exact simulated availability. The Australian simulation constructs a profile-derived proxy and uses it as the primary replay record; neither this proxy nor modeled delivery is a field measurement of curtailed export.

Assumption 1 (Post-event available-export information)**.** *The comparison uses an available-export record a_i^{tel} for the completed interval. Its origin and time alignment must support the delivery calculation under an alternative award. In the controlled study the record is exact simulated availability; in the Australian study the primary record is the profile-derived proxy a_i^{prof}. Record sensitivities perturb the replay input a_i^{tel}, while unperturbed a_i^{prof} continues to generate modeled delivery. Neither record changes an award already issued.*

Assumption 2 (Same-state replay)**.** *Only admitted requests change in the completed-interval comparison. Topology, exogenous network data, participants, rule parameters and underlying available export remain fixed. Awards and power flows are recalculated. If a different award would change availability through inverter controls, storage, demand or portfolio dispatch, those responses must instead enter the replay model.*

4.2 Request choice and delivery displacement

The controlled study links a participant's request to later delivery through one conditional distribution of available export. Participant i observes a private signal s_i, forms a point forecast a_i^{fcst}, and assigns conditional cumulative distribution $F_i(\cdot \mid s_i)$ to its available export. Let $r_i^{\mathrm{ref}} = a_i^{\mathrm{fcst}}$ be its point-forecast request and $r_i^{\mathrm{chg}} = \gamma_i r_i^{\mathrm{ref}}$, $\gamma_i \geq 1$, a larger request. Holding other requests and the network state fixed, the participant values own delivery and faces a modeled shortfall cost $\vartheta \geq 0$ per unit of unused award, normalized by the value of delivered export:

$$U_i(\gamma_i; \vartheta) = \mathbb{E}\big[\min\{x_i(\gamma_i), a_i\} - \vartheta[x_i(\gamma_i) - a_i]^+ \mid s_i\big]\,. \tag{2}$$

Here $[v]^+ = \max\{v, 0\}$ is the positive part. The selected multiplier is the smallest maximizer on the prespecified grid Γ. The same conditional available-export distribution determines this choice and the subsequent realization; the request and delivery curves are not generated independently. A static changed-request case is classified as a request–access–delivery externality if the changed group's expected payoff and unused award both rise while aggregate expected delivery falls beyond the numerical tolerance. A productive high request raises the group's payoff without such a delivery loss. Dynamic cases use realized counterparts. This economic classification is distinct from ECF's operational replay test.

Where the allocation map is differentiable and the available-export CDFs are continuous at the current awards, write $J_{ji} = \partial x_j / \partial r_i$ and $\bar{Y} = \sum_j \mathbb{E}[\min\{x_j, a_j\} \mid s_j]$ for expected total delivery. Here $F_j(x_j)$ abbreviates the conditional cumulative distribution $F_j(x_j \mid s_j)$. Then

$$\begin{aligned} \frac{\partial U_i}{\partial r_i} &= J_{ii}[1 - (1 + \vartheta)F_i(x_i)], \\ \frac{\partial \bar{Y}}{\partial r_i} &= \sum_j J_{ji}[1 - F_j(x_j)]. \end{aligned} \tag{3}$$

The first derivative need not have the sign of the second because the private payoff excludes delivery lost elsewhere. The discrete output tree and changes in active constraints can violate these smoothness conditions. All numerical classifications therefore use matched finite differences. When both quantities are positive, outside-delivery loss divided by the additional unused award gives the diagnostic ratio plotted in Fig. 3(d). Its finite-change definition and relation to the local marginal shortfall cost are in Supplementary Section S2; that section also defines the capacity-normalized Jain index.

4.3 Allocation rules and physical checks

Five rules use the same admitted requests and controlled-study network domain (Supplementary Table 2). Weighted proportional (WP) scales all requests by one feasible factor. Equal reduction (ER) subtracts a common amount; equal access (EA) applies a common cap. Maximum export (ME) maximizes total authorized access. Fairness-aware allocation (FA) trades authorized access against a request-normalized equality term. Their one-parameter formulas, conic objectives, deterministic tie-breaks and sensitivity derivations are specified in Supplementary Sections S5–S8. Section S4 derives the matched request-reduction comparison. All rules allocate against an affine second-order-cone approximation of voltage and two-ended branch limits; the approximation is not treated as the final AC feasibility check.

The controlled medium-voltage case uses the published radial 141-bus network[26], a 0.70 demand scale and 30 fixed connection points. Fixed synthetic branch ratings define this mechanism experiment; they are not native equipment nameplates. Q0 denotes zero participant reactive power and Q95 denotes 0.95 power factor with reactive-power absorption. Each matched reference, changed-request, award, simulated-delivery and replay-delivery state is checked by nonlinear radial AC power flow at its fixed reactive-power setting. The check requires convergence, active- and reactive-power balance, $0.95 \le v_b \le 1.05$ pu, where v_b is bus-voltage magnitude, and

$$\frac{\max_{e\in\{\mathrm{fr,to}\}} |S_{\ell,e}|}{\bar{S}^{\mathrm{syn}}_{\ell}} \le 1 \quad \text{for every monitored branch } \ell. \tag{4}$$

The branch test uses MVA at both ends, independently of rooted flow orientation; $\bar{S}^{\mathrm{syn}}_{\ell}$ is the assigned rating of branch ℓ. Supplementary Section S8 gives the rating construction, calibration, and AC verification details.

4.4 Completed-interval replay and feedback

With unused-award tolerance $\epsilon_x > 0$, let $\mathcal{N}_{\mathrm{ua}}(t) = \{i : x_i(t) - y_i(t) > \epsilon_x\}$. After completion, ECF replaces only this group's admitted requests by the smaller of admission and completed delivery:

$$r^{\mathrm{rpl}}_i(t) = \begin{cases} \min\{r^{\mathrm{adm}}_i(t), y_i(t)\}, & i \in \mathcal{N}_{\mathrm{ua}}(t), \\ r^{\mathrm{adm}}_i(t), & i \notin \mathcal{N}_{\mathrm{ua}}(t). \end{cases} \tag{5}$$

The same allocation rule and network data give replay access $\mathbf{x}^{\mathrm{rpl}}$. Available export then bounds replay delivery:

$$y^{\mathrm{rpl}}_i = \min\{x^{\mathrm{rpl}}_i, a^{\mathrm{tel}}_i\}.$$

Reducing a request to completed delivery does not guarantee that the replay awards that amount. The total-delivery condition below includes any replay loss within the unused-award group. The completed award and delivery are never overridden. Outside the unused-award group, the test sums positive participant-level changes in access and in delivery separately. It also computes the signed change in delivery over all participants. The episode qualifies only if each of these three quantities exceeds τ_{rpl} and the replay passes the applicable nonlinear AC checks. Supplementary Section S3 gives the three expressions. The test qualifies a set-level opportunity; it does not attribute a loss to each member individually. With exact completed delivery and a validated record-error bound $|a^{\mathrm{tel}}_i - a_i| \le \delta_i$, the conservative replay uses

$$\max\{y_i, a^{\mathrm{tel}}_i - \delta_i\}$$

as a lower bound on available export. Supplementary Section S3 proves the resulting lower bound on recovered delivery. Physical checks remain separate. With exact availability and the same recovery threshold, positive aggregate delivery recovery implies positive outside delivery and access recovery. We retain all three quantities to show how recovery occurs and to evaluate perturbed records; they are not three independent filters in the exact-record model.

Let $g(t) = 1$ when all replay conditions hold and zero otherwise. The target and next-interval ceiling are

$$\psi_i^{\mathrm{tar}}(t) = \begin{cases} y_i(t), & g(t) = 1 \text{ and } i \in \mathcal{N}_{\mathrm{ua}}(t), \\ C_i, & \text{otherwise}, \end{cases}$$
$$\psi_i(t+1) = (1 - \beta_{\mathrm{ECF}})\psi_i(t) + \beta_{\mathrm{ECF}}\psi_i^{\mathrm{tar}}(t), \qquad 0 \le \beta_{\mathrm{ECF}} \le 1. \tag{6}$$

For positive gain, only a qualifying episode draws its unused-award group's ceilings toward delivery. Otherwise ceilings recover toward connection bounds. Consecutive nonqualifying intervals give geometric recovery for positive gain; a larger raw request cannot bypass a binding ceiling.

The shortfall-cost benchmark changes request choice but has no feedback state. Mismatch-only feedback updates a smoothed delivery-to-award ratio without replay. Another control applies ECF's update after every unused award without qualification. To test whether a similar update frequency reproduces ECF's advantage, 12 fixed replay-free schedules select unused-award intervals with probability 483/3091, matching ECF's original update fraction among the 3,091 unused-award intervals, not among all 3,840 half-hours. This retrospectively frequency-calibrated comparison tests interval selection at a similar update rate. Each schedule evolves its own awards, delivery and ceilings, so update magnitudes and durations can also differ. Schedules are averaged within each date. Supplementary Section S11.3 gives the selection procedure and all outcomes.

The equal-total control matches ECF's total request reduction in each interval but spreads it proportionally across connections. If $r_i^{\mathrm{cap}} = \min\{r_i, C_i\}$ and $\Delta R_{\mathrm{ECF}} = \sum_i (r_i^{\mathrm{cap}} - r_i^{\mathrm{adm,ECF}})$, the control admits

$$r_i^{\mathrm{adm,uni}} = r_i^{\mathrm{cap}} \left(1 - \frac{\Delta R_{\mathrm{ECF}}}{\sum_j r_j^{\mathrm{cap}}}\right) \tag{7}$$

when the denominator is positive, and retains the capacity-limited request otherwise. This isolates where requests are reduced. With total authorization $X = \sum_i x_i$ and unused authorization $H = \sum_i (x_i - y_i)$, the accounting identity $Y = X - H$ yields

$$Y^{\mathrm{ECF}} - Y^{\mathrm{uni}} = (X^{\mathrm{ECF}} - X^{\mathrm{uni}}) + (H^{\mathrm{uni}} - H^{\mathrm{ECF}}). \tag{8}$$

It does not assert that either term has a fixed sign. Supplementary Section S4 derives the corresponding redistribution conditions.

4.5 Controlled experiments

A five-node conditional available-export tree uses standardized support $(-2, -1, 0, 1, 2)$ with probabilities $(1, 4, 6, 4, 1)/16$ and scale σ. The participant's forecast anchors both its request and the later node realization. The static grid spans 30 locations, five rules, Q0 and Q95, five uncertainty levels and nine shortfall-cost levels, giving 13,500 request-choice cases. The selected payoff-maximizing multiplier is held fixed within a 16-interval trajectory. Three temporal profiles have identical individual node counts but differ in ordering: clustered, intermittent and shock–recovery. Participant-specific

Table 2: How the comparison mechanisms differ. C denotes the controlled study and A the Australian simulation. Paired controls use the same network and available-export trajectory.

Mechanism	What changes	Question tested
No feedback (C, A)	Ceiling stays at the connection bound.	Baseline delivery.
Shortfall cost (C)	A modeled cost changes the initial request.	Effect of an incentive before allocation.
Mismatch-only feedback (C)	A participant's own delivery-to-award ratio updates its ceiling.	Effect of responding to shortfall alone.
Update without replay test (A)	ECF's formula applies after every unused award.	Effect of removing the test, including more frequent updates.
Frequency-calibrated random update (A)	The same formula updates randomly selected unused-award intervals.	Whether a similar update frequency explains ECF's gain.
Equal-total reduction (C, A)	ECF's total reduction is spread proportionally.	Value of where requests are reduced.
Previous-day persistence (A)	Request uses the previous-day output estimate alone.	Cost of a conservative request.
ECF (C, A)	Qualifying unused awards lower later ceilings.	Combined effect of testing and updating.

phase offsets also change the simultaneous combinations of nodes across connections. Changed requests come from one participant, five electrically nearby participants, or five dispersed participants. In the five-participant cases, each uses its own best-response multiplier and all five requests change simultaneously. The 270 base settings, each evaluated under six mechanisms, give 1,620 trajectories and 25,920 interval comparisons. Supplementary Table 3 gives every factor, numerical tolerance and parameter level. The post-event record is exact simulated availability in this controlled study.

4.6 Australian public-data simulation

The external simulation combines a published unbalanced 11-kV feeder with 1,255 service connections and its LV sections[27] with half-hour PV and consumption profiles from 300 Ausgrid solar homes[28]. These sources have no common customer identifier. The primary deterministic profile-to-service mapping is fixed before assessment, and separate placements test its sensitivity. The forward-uptake scenario assigns 418 connections export participation and scales PV to an approximately 9.9-kWp mean[29]. It is not a historical reconstruction of issued operating envelopes. The day-to-day forecast increment is estimated from 2010–2011 data; 2011–2012 data select fixed feeder settings; 2012–2013 provide the assessment chronology. A day is eligible only when mapped profiles are complete on that day and the previous day. Excluding missing and daylight-saving-transition records leaves 80 network days, or 3,840 half-hours per control.

For mapped profile $\iota(i)$, source gross PV generation GG, general consumption GC and controlled load CL are first converted to interval-average MW. Applying the scenario multipliers gives

$$a_i^{\mathrm{prof}}(d,\tau) = \left[\lambda_{\mathrm{PV}}\mathrm{GG}_{\iota(i)}(d,\tau) - \lambda_L(\mathrm{GC}_{\iota(i)}(d,\tau) + \mathrm{CL}_{\iota(i)}(d,\tau))\right]^+ . \tag{9}$$

Modeled delivery under control c is $y_i^c = \min\{x_i^c, a_i^{\mathrm{prof}}\}$; the unperturbed profile proxy also supplies the primary replay record. Raw requests use the previous day's same-half-hour proxy plus a positive 90th-percentile increment from same-half-hour day-to-day changes. Contemporaneous household

demand is fixed as an exogenous input to this retrospective allocation; load-forecast error is not modeled.

For AC delivery calculations, household consumption is first offset by local PV; the remaining net import minus delivered net export is the active-power load. The authorization screen instead sets participant net injection equal to the full award, without using realized PV to relax the constraint. Household reactive demand is retained in both screens. Supplementary Section S10.4 gives the connection-level equations and mapping.

The primary feeder settings use a 0.30 load multiplier and a 1.07-pu allocation-stage upper-voltage limit, below the 1.10-pu physical limit. For each half-hour, a zero-anchored feasible-factor search scales admitted requests under nonlinear unbalanced AC power flow. An independent factor-grid check of all 19,200 primary allocation calls found no feasible sampled factor above the returned one; Supplementary Section S10.4 states the search resolution and its limits. Load-only, authorized-export and modeled-delivery states are screened separately for convergence, customer voltage and native transformer ratings. Because the public model lacks usable native line ampacities, line-thermal certification is outside this physical screen. The replay threshold is 2 W, a numerical decision threshold rather than an instrument-accuracy claim.

Each primary day begins with ceilings at C_i and compares no feedback, previous-day persistence, ECF at $\beta_{\mathrm{ECF}} = 0.5$, and equal-total reduction. The additional comparisons vary gain, replay qualification, cross-day memory, record perturbations and error bounds, profile placement, PV capacity, participant adaptation and published LV networks while retaining the documented fixed inputs for each matched comparison. Source-terminal power from the same AC solves accounts for upstream exchange. The request quantile and fixed feeder settings are calculated from the earlier source years. Initial exploratory runs also informed the operating setting. All reported 80-day comparisons use the resulting common setting. Supplementary Sections S10.3–S10.4 give the construction sequence and full model settings.

4.7 Statistical analysis

The primary comparison unit in the Australian study is one paired network day. A day contains 48 half-hours on the same network and profile mapping; participants, half-hours and nested LV sections are not treated as independent network samples. Energy differences are computed within each day before averaging. We report ordinary paired-day percentile intervals and a date-dependent wild-bootstrap interval that allows short-range correlation between nearby eligible dates. Both use 10,000 replicates. The request- adaptation experiment averages three seeds within each date; the frequency-calibrated-update comparison averages 12 schedules within each date. Intervals are pointwise and do not adjust for multiple comparisons. Supplementary Section S10.3 gives the calendar covariance, exact eligibility rules and full numerical settings.

Author contributions

Junjie Huang: conceptualization, methodology, software, formal analysis, investigation, visualization, data curation, and writing—original draft. Tao Yu and Yusheng Xue: conceptualization, supervision, funding acquisition, and writing—review and editing. Zhenning Pan, Yufeng Wu, and Wencong Xiao: methodology, validation, and writing—review and editing. Rankai Zhu: validation, data curation, and writing—review and editing. Jie Huang: validation and writing—review and editing.

Competing interests

The authors declare no competing interests.

Data availability

Calculated results, figure source data, input-source identifiers, preprocessing instructions, date selections and network-profile mapping tables are available in the public project repository, `https://github.com/MarxHuang/ecf-export-access`. The source 141-bus network, CSIRO/GridQube network collection (doi:10.25919/ghnz-bk28) and Ausgrid Solar Home Electricity Data remain with their original distributors under their respective terms; raw third-party records are not redistributed. The repository distinguishes deposited calculated data from those external inputs required for a full simulation rerun.

Code availability

Simulation, preprocessing and analysis code, parameter files, environment specifications and reproduction instructions are available at `https://github.com/MarxHuang/ecf-export-access`. The repository identifies its deposited revision and provides checks of saved results separately from complete reruns, which require the source datasets described above.

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